MA3521 — Eigenvalue Systems

Name: Answer key

Problem 1 Find the GS, then use the IC to determine the constants

$$\mathbf{y}' = A\mathbf{y} \qquad A = \begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix} \qquad \mathbf{y}(0) = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$

1a. Eigenvalues and eigenvectors

$\det(A - \lambda I) =$ $(2-\lambda)(2-\lambda) - 9$ $= 0$

$\det(A-\lambda I) = 0$

$\begin{vmatrix} 2-\lambda & 3 \\ 3 & 2-\lambda \end{vmatrix} = 0$

$(2-\lambda)(2-\lambda) - 9 = 0$

$4 - 4\lambda + \lambda^2 - 9 = 0$

$\lambda^2 - 4\lambda - 5 = 0$

$(\lambda - 5)(\lambda + 1) = 0$

$\lambda_1 = 5$

$\lambda_2 = -1$

$\lambda_1 = 5$ $\quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$\lambda_2 = -1$ $\quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\lambda_1 = 5$

$A - 5I = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix}$

$\begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$-3x + 3y = 0 \quad x = y$

$3x - 3y = 0 \quad x = 1 \Rightarrow y = 1$

$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$\lambda_2 = -1$

$A + 1I = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix}$

$\begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$3x + 3y = 0$

$y = -x$

$x = 1, \; y = -1$

$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Choose $x = 1$ for simplicity

1b. General solution

$\mathbf{y}_c(t) =$ $c_1 e^{5t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

general sol:

$y_c(t) = c_1 e^{\lambda_1 t} \mathbf{v}_1 + c_2 e^{\lambda_2 t} \mathbf{v}_2$

$y_c(t) = c_1 e^{5t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Write down $B\mathbf{c} = \mathbf{b}$ to solve for $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

$B =$ $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ $\quad \mathbf{b} =$ $\begin{pmatrix} 2 \\ 0 \end{pmatrix}$

$B\mathbf{c} = \mathbf{b}$

$B = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \quad \mathbf{b} = \begin{pmatrix} 2 \\ 0 \end{pmatrix}$ ← initial cond.

$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$

$c_1 + c_2 = 2$

$c_1 - c_2 = 0$

$2c_1 = 2$

$c_1 = 1$

$1 + c_2 = 2$

$c_2 = 1$

$y(t) = e^{5t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + e^{-t} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$y(t) = \begin{bmatrix} e^{5t} + e^{-t} \\ e^{5t} - e^{-t} \end{bmatrix}$

$\mathbf{y}(t) =$ $\begin{bmatrix} e^{5t} + e^{-t} \\ e^{5t} - e^{-t} \end{bmatrix}$ as $t \to \infty$ $\sim$ $e^{5t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Note: as $t \to \infty$

$e^{5t} \to \infty$

$e^{-t} \to 0$

so $y(t) \sim \begin{bmatrix} e^{5t} + 0 \\ e^{5t} + 0 \end{bmatrix}$

OR $e^{5t} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Problem 2 Find the GS, then use the IC to determine the constants

$$\mathbf{y}' = A\mathbf{y} \qquad A = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix} \qquad \mathbf{y}(0) = \begin{pmatrix} 1 \\ 4 \end{pmatrix}$$

2a. Eigenvalues and eigenvectors

$\det(A - \lambda I) =$ $(1-\lambda)(2-\lambda)-6$ $= 0$

$\det(A-\lambda I)=0$

$\begin{vmatrix} 1-\lambda & 2 \\ 3 & 2-\lambda \end{vmatrix} = 0$

$(1-\lambda)(2-\lambda)-6=0$

$2-3\lambda+\lambda^2-6=0$

$\lambda^2-3\lambda-4=0$

$(\lambda-4)(\lambda+1)=0$

$\lambda_1=4, \lambda_2=-1$

$\lambda_1 =$ 4 $\quad \mathbf{v}_1 =$ $\begin{pmatrix} 2 \\ 3 \end{pmatrix}$

$\lambda_2 =$ -1 $\quad \mathbf{v}_2 =$ $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

$\lambda_1=4$

$A-4I = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$

$= \begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix}$

$\begin{bmatrix} -3 & 2 \\ 3 & -2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$-3x+2y=0$

$y=\frac{3}{2}x$

$x=2$, $y=3$ $\quad v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$\lambda_2=-1$

$A+1I = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

$= \begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix}$

$\begin{bmatrix} 2 & 2 \\ 3 & 3 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$2x+2y=0$

$y=-x$

$x=1$, $y=-1$ $\quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

choose $x=1$ for simplicity

general solution:

$y_c(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2$

$y_c(t) = c_1 e^{4t}\begin{bmatrix} 2 \\ 3 \end{bmatrix} + c_2 e^{-t}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

2b. General solution

$\mathbf{y}_c(t) =$ $c_1 e^{4t}\begin{bmatrix} 2 \\ 3 \end{bmatrix} + c_2 e^{-t}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$Bc = b$

$B = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$ $\quad b = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$ $\rightarrow \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

$2c_1 + c_2 = 1$

$3c_1 - c_2 = 4$

$5c_1 = 5$

$c_1 = 1$ $\qquad 2(1) + c_2 = 1$

$c_2 = -1$

Write down $B\mathbf{c} = \mathbf{b}$ to solve for $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

$y(t) = e^{4t}\begin{bmatrix} 2 \\ 3 \end{bmatrix} - e^{-t}\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$= \begin{bmatrix} 2e^{4t} - e^{-t} \\ 3e^{4t} + e^{-t} \end{bmatrix}$

$B =$ $\begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$ $\quad \mathbf{b} =$ $\begin{pmatrix} 1 \\ 4 \end{pmatrix}$

$\mathbf{y}(t) =$ $\begin{bmatrix} 2e^{4t} - e^{-t} \\ 3e^{4t} + e^{-t} \end{bmatrix}$ as $t \to \infty$ $\sim$ $e^{4t}\begin{bmatrix} 2 \\ 3 \end{bmatrix}$

Note: as $t \to \infty$

$e^{4t} \to \infty$

$e^{-t} \to 0$

$y(t) \sim \begin{bmatrix} 2e^{4t} - 0 \\ 3e^{4t} + 0 \end{bmatrix}$

$\sim e^{4t}\begin{bmatrix} 2 \\ 3 \end{bmatrix}$